

Lab Task 03: Using CrypTool 2

Objective:

To practice encryption, decryption, and cryptanalysis of the Vigenère cipher using CrypTool 2.

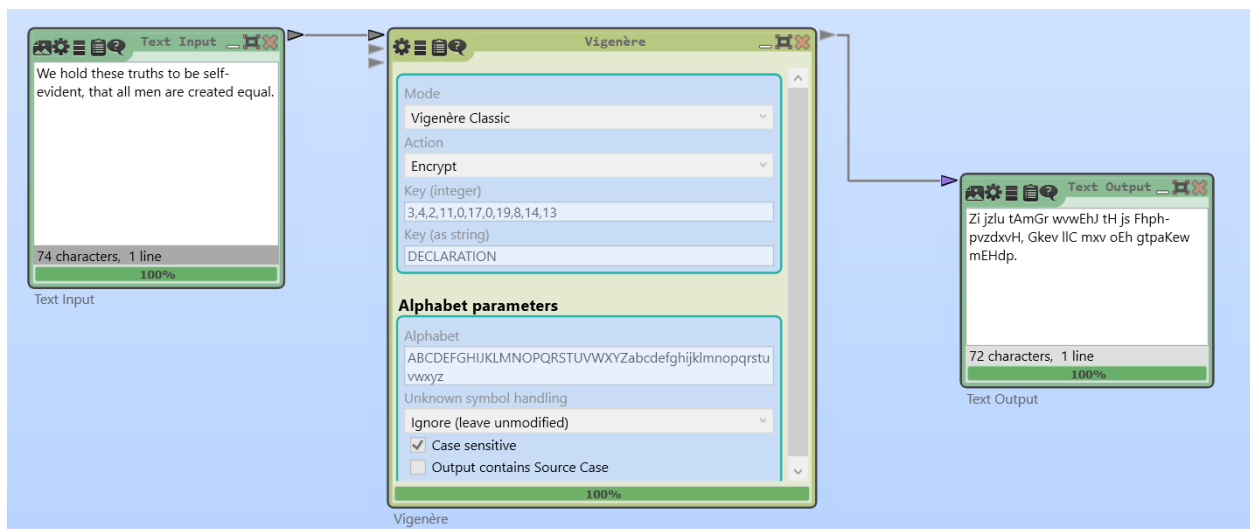
Procedure:

- Open CrypTool 2 and select the Vigenère cipher from the classical ciphers toolbox.
- Enter the plaintext and the key (example: DECLARATION).
- Run the tool to encrypt the message and record the ciphertext.
- Use CrypTool's analysis functions (Kasiski, Friedman, frequency analysis).
- Attempt to recover the key and decrypt the ciphertext.
- Optionally, perform a known-plaintext attack for verification.

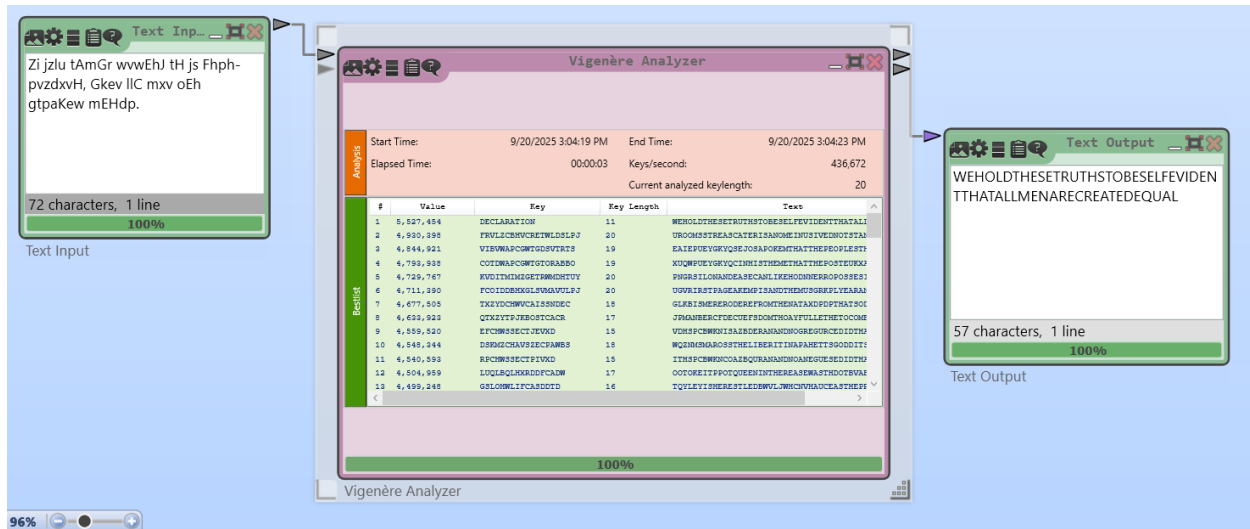
Results:

(a) Vigenère Cipher Analysis

Vigenère Encryption Flow:

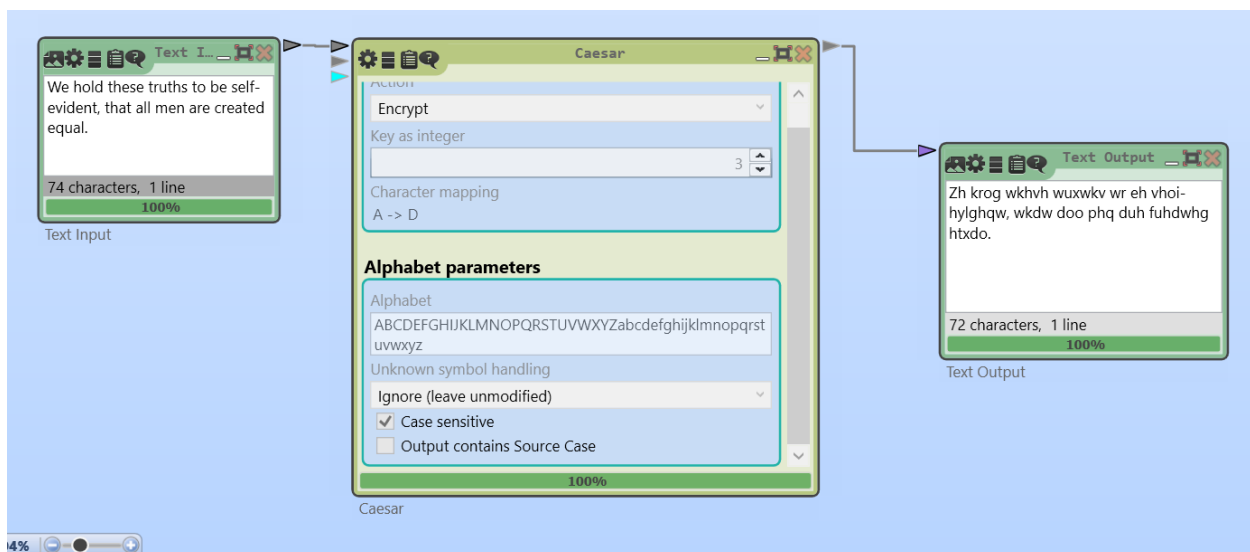


Decryption (using Vigenère Analysis)



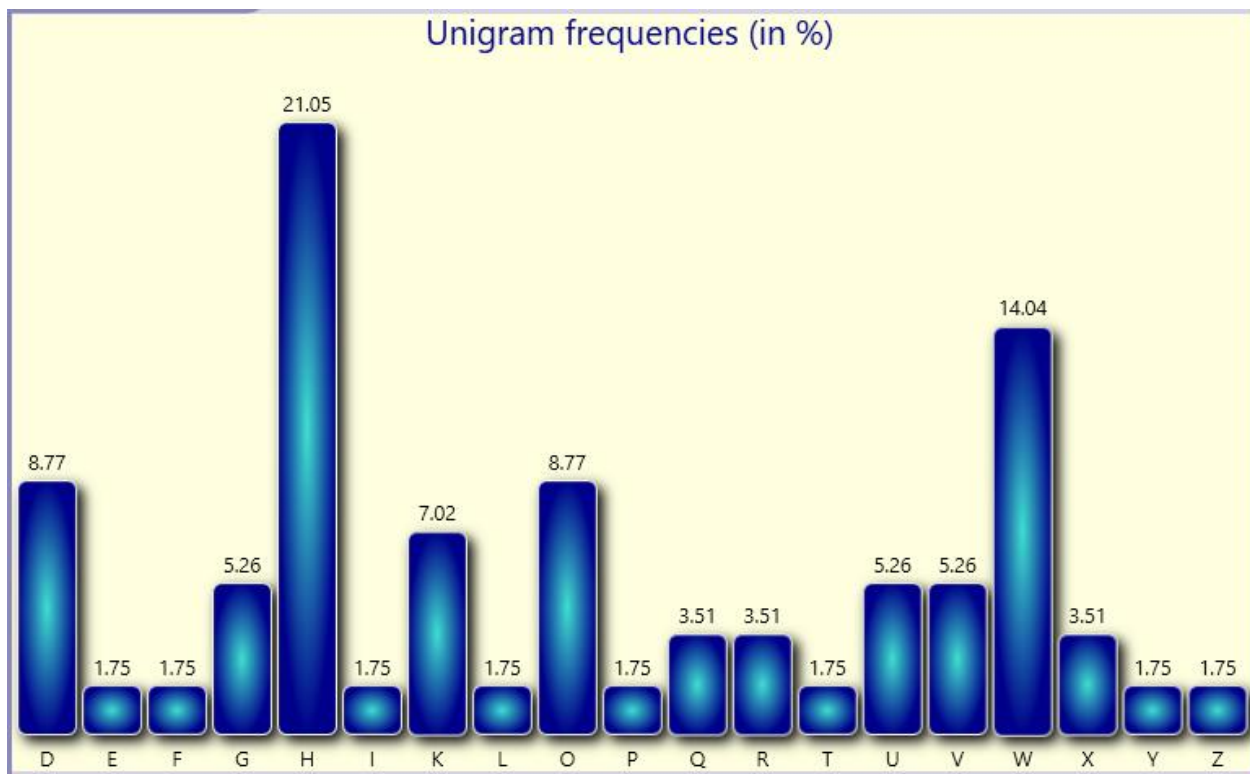
(b) Frequency Analysis

Encrypted using Caesar Cipher:

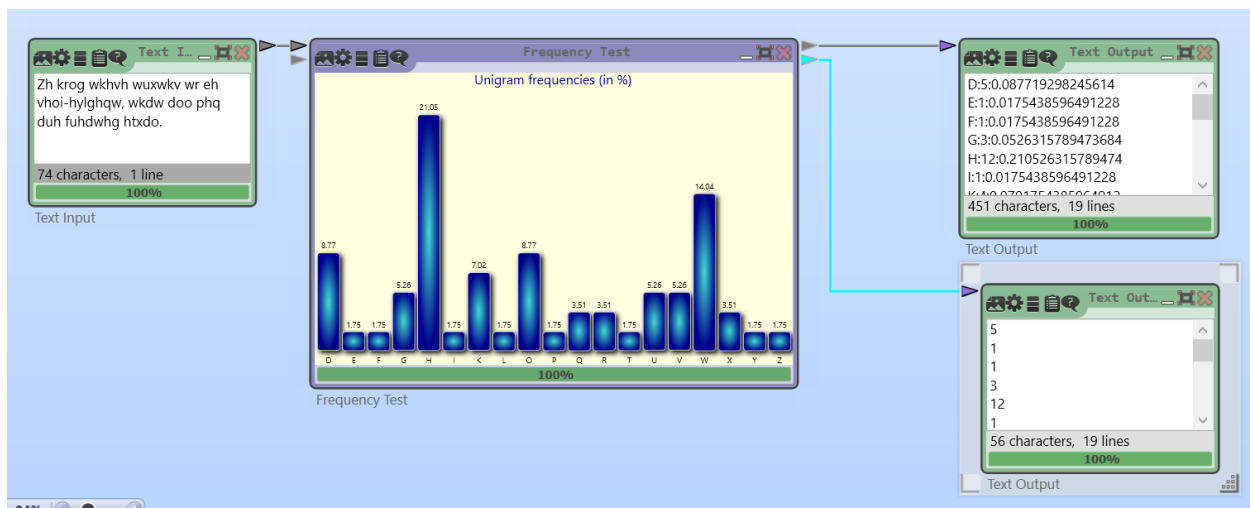


Ciphertext generated for sample plaintext and key

Frequency Analysis chart obtained:



Frequency Analysis Flow:



Frequency Analysis only provides hints or clues about the possible plaintext. To determine the exact keys, we still need manual observation and adjustments.